

Day, Night & Seasons

Use the cue, then fade the cue. Study the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

CUE FOCUS Earth's **rotation** means it spins around once about every **24 hours**. This spinning causes **day and night**. Earth's **revolution** means it orbits the Sun once a **year**. Because Earth is **tilted**, its revolution causes the **seasons** and the changing amount of daylight.

Round 1 — Retrieve It

1. What is it called when Earth spins around once?

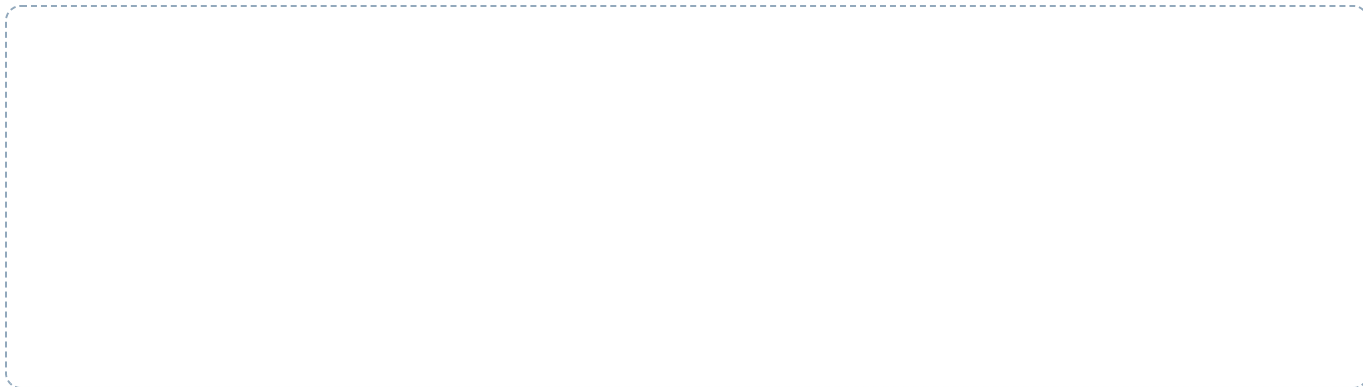
2. About how long does one rotation of Earth take?

3. What does Earth's rotation cause?

4. What causes the seasons to change?

Round 2 — Sketch It

Draw the Sun and Earth. Show the side of Earth facing the Sun as day and the side facing away as night. Add a curved arrow to show Earth spinning (rotation).



Day, Night & Seasons — Apply It

Keep the cue card covered. Answer first. Then use the cue to check, revise, and explain what you changed.

Round 3 — Apply the Idea

1. It is daytime for you but nighttime for a friend across the world. Which motion of Earth explains this, and why?

2. One trip around the Sun takes about a year. Which motion is this, and what does it help cause?

3. In summer there are more hours of daylight than in winter. What causes this change in daylight?

Round 4 — Reason from Evidence

A student says, "It is summer because Earth is closer to the Sun." Using rotation, revolution, and tilt, explain what really causes the seasons.

Day, Night & Seasons — ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain the difference between rotation and revolution. Tell what each one causes.

C**Connect**

Connect Earth's tilt to the seasons. Why does the tilt matter as Earth travels around the Sun?

E**Extend**

If Earth spun twice as fast, predict how long a day would be. Explain your thinking.

Confidence check. Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it.

Teacher Key — Day, Night & Seasons

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. Rotation.
2. About 24 hours (one day).
3. Day and night.
4. Earth's revolution around the Sun combined with Earth's tilt.

Sketch It — look for

Sun and Earth drawn; the side facing the Sun labeled day and the side facing away labeled night, with a curved arrow showing rotation (spinning).

Apply It

1. Rotation; as Earth spins, one side faces the Sun (day) while the other faces away (night).
2. Revolution; it helps cause the seasons and the change in daylight (along with Earth's tilt).
3. Earth's tilt as it revolves around the Sun; the part tilted toward the Sun gets more daylight.

Reason from Evidence — look for

Distance is not the reason. Rotation causes day and night. Revolution plus Earth's tilt causes the seasons: the hemisphere tilted toward the Sun gets more direct light and longer days, making it warmer (summer).

ACE — Extend look-for

A day would be about half as long (roughly 12 hours) because rotation causes day and night, so faster spinning means shorter days.

Suggested use. Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color.